

EX.NO:11	FILE INFORMATION CHECKER
DATE:	

AIM:

To write a Java program to check if a file exists, is readable, writable, determine its type, and display its size.

ALGORITHM:

Step-1: Start the program.

Step-2: Accept a file name from the user.

Step-3: Create a File object using the file name.

Step-4: Check if the file exists using exists().

Step-5: Check if the file is readable using canRead().

Step-6: Check if the file is writable using canWrite().

Step-7: Determine if the file is a file or folder using isFile().

Step-8: If it exists, get the file size using length().

Step-9: Display all information.

Step-10: End the program.

PROGRAM:

```
import java.io.File;
import java.util.Scanner;
public class FileInfo {
public static void main(String[] args) {
Scanner sc = new Scanner(System.in);
System.out.print("Enter filename:");
File f = new File(sc.nextLine());
System.out.println("Exists: "+f.exists());
System.out.println("Can Read: "+f.canRead());
System.out.println("Can Write: "+f.canWrite());
System.out.println("Type: "+(f.isFile()?"File":"Folder"));
if(f.exists()) System.out.println("Size: "+f.length()+" bytes");
}
}
```

OUTPUT:

Enter filename: File Info
Exists: false
Can Read: false
Can Write: false
Type: Folder

RESULT:

Thus the program has been executed successfully.

EX.NO:12	TEXT STYLE CHANGER
DATE:	

AIM:

To write a Java program to change the font size and style of text using GUI controls.

ALGORITHM:

Step-1: Start the program.

Step-2: Create a JFrame for GUI.

Step-3: Add a JTextArea to display text.

Step-4: Add a JComboBox or controls to select font size/style.

Step-5: Apply the selected font size and style to the text area.

Step-6: Display the GUI.

Step-7: End the program.

PROGRAM:

```
import javax.swing.*;

import java.awt.*;

public class TextStyle extends JFrame { public TextStyle() {

setSize(300,200);

setLayout(null);

JTextArea ta = new JTextArea("Hello BCA");

ta.setBounds(10,10,250,100);

JComboBox<String> size = new JComboBox<>(new

String[]{"Small","Medium","Large"});

size.setBounds(10,120,100,30);

add(ta);

add(size);

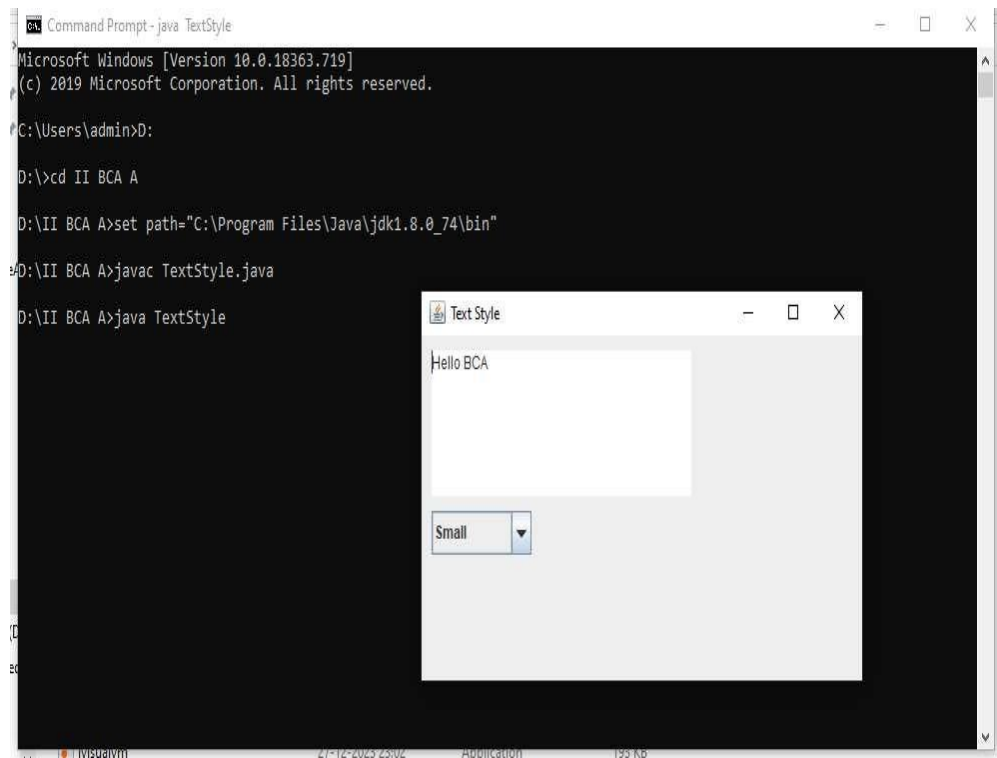
setVisible(true);

}

public static void main(String[] args) { newTextStyle(); }

}
```

OUTPUT:



The screenshot shows a Windows Command Prompt window titled "Command Prompt - java TextStyle" with a black background. The text inside the prompt shows the following commands and their outputs:

```
Microsoft Windows [Version 10.0.18363.719]
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C:\Users\admin>D:

D:\>cd II BCA A

D:\II BCA A>set path="C:\Program Files\Java\jdk1.8.0_74\bin"

D:\II BCA A>javac TextStyle.java

D:\II BCA A>java TextStyle
```

Overlaid on the Command Prompt is a smaller window titled "TextStyle". This window contains a text area with the text "Hello BCA" and a dropdown menu below it with the text "Small".

RESULT:

Thus the program has been executed successfully.

EX.NO:13	MOUSE EVENT DEMO
DATE:	

AIM:

To write a Java program to demonstrate mouse events using Adapter classes and display the event information.

ALGORITHM:

Step-1: Start the program.

Step-2: Create a JFrame for GUI.

Step-3: Add a JLabel to display messages.

Step-4: Use MouseAdapter to handle mouse events.

Step-5: On mouse events (click, enter, exit), update the label.

Step-6: Display the GUI.

Step-7: End the program.

PROGRAM:

```
import javax.swing.*;
import java.awt.event.*;
public class MouseDemo extends JFrame {
    JLabel    msg = new JLabel("Click anywhere!");
    public MouseDemo() {
        setSize(300,200);
        setLayout(null);
        msg.setBounds(100,80,200,30);
        addMouseListener(new MouseAdapter() {
            public void mouseClicked(MouseEvent e)
            {
                msg.setText("Clicked at (" + e.getX() + ", " + e.getY() + ")");
            }
        });
        add(msg);
        setVisible(true);
    }
    public static void main(String[] args) { new MouseDemo(); }
}
```

OUTPUT:

```
Microsoft Windows [Version 10.0.18363.719]
(c) 2019 Microsoft Corporation. All rights reserved.

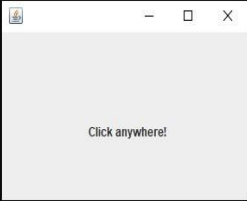
C:\Users\admin>d:

D:\>cd II BCA A

D:\II BCA A>SET PATH="C:\Program Files\Java\jdk1.8.0_74\bin"

D:\II BCA A>javac MouseDemo.java

D:\II BCA A>java MouseDemo
```



```
Microsoft Windows [Version 10.0.18363.719]
(c) 2019 Microsoft Corporation. All rights reserved.

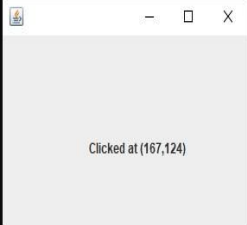
C:\Users\admin>d:

D:\>cd II BCA A

D:\II BCA A>SET PATH="C:\Program Files\Java\jdk1.8.0_74\bin"

D:\II BCA A>javac MouseDemo.java

D:\II BCA A>java MouseDemo
```



RESULT:

Thus the program has been executed successfully.

EX.NO:14	BASIC CALCULATOR
DATE:	

AIM:

To write a Java program to create a simple calculator with basic arithmetic operations using GUI.

ALGORITHM:

Step-1: Start the program.

Step-2: Create a JFrame and JTextField for display.

Step-3: Add buttons for digits and operators in a GridLayout.

Step-4: On button click, update the display or perform the calculation.

Step-5: Handle divide by zero or invalid input exceptions.

Step-6: Display the result in the text field.

Step-7: End the program.

PROGRAM:

```
import javax.swing.*;
import java.awt.*;
import java.awt.event.*;

public class StudentCalc extends JFrame implements ActionListener {
    JTextField display = new JTextField("0");
    int num1 = 0;
    String operator = "";

    StudentCalc() {
        display.setFont(new Font("Arial", Font.BOLD, 18));
        display.setHorizontalAlignment(JTextField.RIGHT);
        display.setEditable(false);
        add(display, BorderLayout.NORTH);

        JPanel p = new JPanel(new GridLayout(4,3)); String[] buttons = {
            "1","2","3",
            "4","5","6",
            "7","8","9",
            "+","0","="
        };

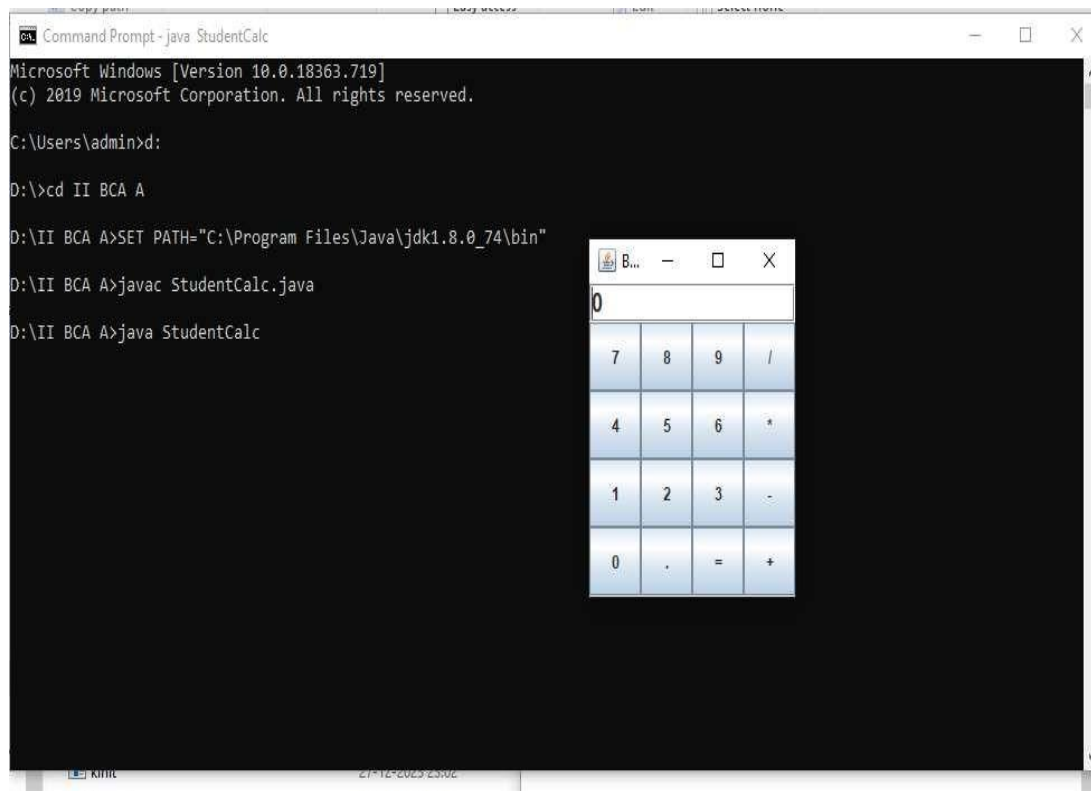
        for (String s : buttons)
        { JButton b = new JButton(s);
          b.addActionListener(this);
          p.add(b);
        }
        add(p);
        setTitle("Simple Calc");
        setSize(200,250);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setVisible(true);

        public void actionPerformed(ActionEvent e) {
```

```
String key = e.getActionCommand();
if ("0123456789".contains(key))
{ if (display.getText().equals("0"))
display.setText(key); else
display.setText(display.getText() + key);
}
else if (key.equals("+")) {
num1 = Integer.parseInt(display.getText());
operator = "+";
display.setText("0");
}
else if (key.equals("=")) {
int num2 = Integer.parseInt(display.getText());
result = 0;
if (operator.equals("+")) result = num1 + num2;
display.setText("" + result);
}
}

public static void main(String[] args) { new StudentCalc();
}
}
```

OUTPUT:



```
Command Prompt - java StudentCalc
Microsoft Windows [Version 10.0.18363.719]
(c) 2019 Microsoft Corporation. All rights reserved.

C:\Users\admin>:

D:\>cd II BCA A

D:\II BCA A>SET PATH="C:\Program Files\Java\jdk1.8.0_74\bin"

D:\II BCA A>javac StudentCalc.java

D:\II BCA A>java StudentCalc
```

RESULT:

Thus the program has been executed successfully.

EX.NO:15	TRAFFIC LIGHT SIMULATOR
DATE:	

AIM:

To write a Java program to simulate a traffic light using radio buttons and display messages for stop, ready, and go.

ALGORITHM:

Step-1: Start the program.

Step-2: Create a JFrame window to hold the components.

Step-3: Add a JLabel at the top to display the traffic light status.

Step-4: Create three JRadioButtons labeled **RED**, **YELLOW**, and **GREEN**.

Step-5: Group the radio buttons using ButtonGroup so only one can be selected at a time.

Step-6: Add ActionListeners to each radio button:

- RED → set label text to "STOP" and color to RED.
- YELLOW → set label text to "READY" and color to ORANGE.
- GREEN → set label text to "GO" and color to GREEN.

Step-7: Add the radio buttons to a JPanel and add the label and panel to the frame.

Step-8: Set frame properties: title, size, default close operation, center the window, and make it visible.

Step-9: Wait for user interaction; update label text and color when a button is clicked.

Step-10: End the program when the user closes the window.

PROGRAM:

```
import javax.swing.*;
import java.awt.*;

public class TrafficLight extends JFrame {
    JLabel status = new JLabel("Choose Light");

    public TrafficLight() {
        setTitle("Traffic Light");
        setSize(250, 200);
        setLayout(new BorderLayout());
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        setLocationRelativeTo(null); // center window
        status.setFont(new Font("Arial", Font.BOLD, 20));
        status.setHorizontalAlignment(JLabel.CENTER);
        JRadioButton red = new JRadioButton("RED");
        JRadioButton yellow = new JRadioButton("YELLOW");
        JRadioButton green = new JRadioButton("GREEN");
        // Group radio buttons
        ButtonGroup group = new ButtonGroup();
        group.add(red);
        group.add(yellow);
        group.add(green);
        red.addActionListener(e -> {
            status.setText("STOP");
            status.setForeground(Color.RED);
        });
        yellow.addActionListener(e -> {
            status.setText("READY");
            status.setForeground(Color.ORANGE);
        });
        green.addActionListener(e -> {
```

```
status.setText("GO");
status.setForeground(Color.GREEN);
});
JPanel btns = new JPanel();
btns.add(red);
btns.add(yellow);
btns.add(green);
add(status, BorderLayout.NORTH);
add(btns, BorderLayout.SOUTH);
setVisible(true);
}
public static void main(String[] args) { SwingUtilities.invokeLater(TrafficLight::new);
}
}
```

OUTPUT:

```
Command Prompt - java TrafficLight
Microsoft Windows [Version 10.0.26100.7623]
(c) Microsoft Corporation. All rights reserved.

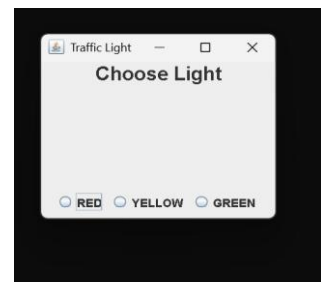
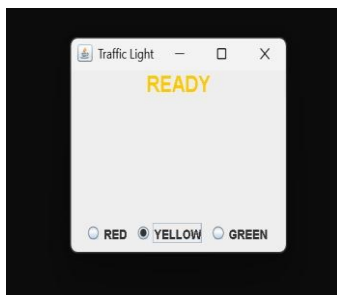
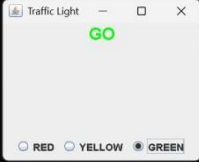
C:\Users\mailt>D:

D:\>CD II BCA A

D:\II BCA A>set path="C:\Program Files\Java\jdk-13.0.1\bin"

D:\II BCA A>javac TrafficLight.java

D:\II BCA A>java TrafficLight
```



RESULT:

Thus the program has been executed successfully

